

Seunghyun Kim

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Education

Daejeon University, College of Engineering

Daejeon, S.Korea

B.S. IN ELECTRONICS, INFORMATION & COMMUNICATIONS ENGINEERING

Mar. 2019 – Feb. 2026 (Expected)

- Double Major: Next-Generation Communication Convergence, DSC Platform University
- GPA: **4.47 / 4.5** (Ranked **1st** out of 90 students)
- Military service: ROK Army (Signal Corps), Oct. 2019 – May. 2021
- Technical Skills: Python, C, PyTorch, TensorFlow, OpenCV
- Selected Courseworks: Computer Networks, Deep Learning & Computer Vision, Applied AI for Autonomous Driving

Publications

[1] “Classification of accident risk situations based on vehicle speed estimation through lane-specific coordinate transformation.” **Seunghyun Kim**, Hanbin Kang, Jonggyu Kang, and Changseok Bae. Journal of Korean Institute of Next Generation Computing, Vol. 21, No. 1, pp. 7–27; Feb. 2025.

[2] “Real-time object detection service of drone with Electro-Optics/Infra-Red image sensors.” **Seunghyun Kim**, Byungbog Lee. In Proceedings of the 2024 Fall Conference of IEMEK, pp. 300–302; Nov. 2024.

Research Experience

Image and Intelligent Information Lab, Daejeon University

Daejeon, S. Korea

UNDERGRADUATE STUDENT RESEARCHER with Prof. Changseok Bae

Jun. 2023 – Feb. 2025

- Proposed an **accident risk classification model** using lane-specific vehicle speed estimation, resulting in a **1st-author** journal publication [1].
- Designed and maintained a **\$150K experimental testbed** (track, robotic car, communication env, server) to collect and label driving datasets.
- Achieved **98% mAP and mAR** in object detection by developing a **custom YOLOv8** model with dataset separation and augmentation.
- Increased **accuracy by 35%** and **inference speed by 224×** over baseline by developing an **MLP-based classifier** integrated with YOLO.
- Reduced speed estimation **error to ±5 km/h** in real-road conditions by applying lane-specific coordinate transformation and calibration.

DNA+Drone Platform Research Center, ETRI

Daejeon, S. Korea

RESEARCH INTERN with Dr. Byungbog Lee

Jul. 2024 – Dec. 2024

- Achieved **0.6 s end-to-end latency** by developing a real-time **EO/IR object detection system** on a drone within a private **5G NR n79 network**.
- Applied slicing, linear interpolation padding as **pre/post-processing** of data in Ubuntu and using TensorRT, **reduced inference time by 0.2s**.
- Boosted **detection accuracy by 8%** through **EO-IR image fusion**, enabling reliable performance in both day and night conditions.
- Received the **Best Paper Presentation Award** at the IEMEK Fall Conference for presenting methodology and results of the detection system[2].

Awards & Scholarships

AWARDS

- | | | |
|------|---|------------------|
| 2024 | Best Paper Presentation Award , 2024 Fall Conference, IEMEK (Institute of Embedded Engineering of Korea) | Jeju, S.Korea |
| 2024 | Grand Prize (Top 1/30+, \$700) , Capstone Design Competition, Dept. of EICE, Daejeon University | Daejeon, S.Korea |
| 2024 | Grand Prize (Top 1/15+, \$1,100) , AIoT Service Design Competition, Daejeon University | Daejeon, S.Korea |
| 2024 | Finalist (\$300) , Hanium ICT Project Contest, FKII (The Federation of Korean Information Industries) | Seoul, S.Korea |

SCHOLARSHIPS

- | | | |
|--------|---|------------------|
| 2019 – | Academic Excellence Scholarship (Top 1, awarded 5 times, \$15,000) , Dept. of Electronic Information and | Daejeon, S.Korea |
| 2024 | Communication Engineering, Daejeon University | |
| 2019 – | Academic Achievement Scholarship (Top 2, awarded 2 times, \$3,000) , Dept. of Electronic Information | Daejeon, S.Korea |
| 2024 | and Communication Engineering, Daejeon University | |
| 2023 – | DSC Scholarship for Innovative Talents (Awarded 4 times, \$6,000) , DSC Platform University | Daejeon, S.Korea |
| 2024 | | |

Extracurricular Activity

HuMechanic (Robotics Club of Daejeon University)

Daejeon, S. Korea

TECHNICAL LEAD & PRESIDENT AT 2023

Mar. 2022 – Feb. 2024

- Led a 20+ member team to develop **lane detection** and path control in Python using Canny edge detection and Hough transform on live camera
- Designed a **YOLOv4-tiny pedestrian detection model** optimized for real-time inference on low-power embedded hardware.
- Organized and delivered **image processing workshops** covering OpenCV, object detection, and real-time computer vision.

International Training Program, UNLV (University of Nevada, Las Vegas)

Las Vegas, USA

PARTICIPANT

Jan. 2024

- Authored and presented a technical report **comparing Camera, LiDAR, and Radar sensors** for autonomous vehicle perception systems.
- Built an **SSD-MobileNet object detection model** with TensorFlow, achieving real-time inference and visualization on a webcam stream.
- Created and optimized an **LSTM-based NMT model** with TensorFlow/Keras, achieving **BLEU-1 of 0.62** and deploying it on Raspberry Pi.